

SANGHOON JI

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RESEARCH INTERESTS

GIS & Geomatics; Spatial Networks & Infrastructure Accessibility; Ecosystem Monitoring; Machine Learning; Spatiotemporal Data Analysis; Remote Sensing & 3D Reconstruction

EDUCATION

KYUNG HEE UNIVERSITY

Seoul, Korea

M.S. in Geography (GPA 4.38/4.5), Advisor: Dr. Chulsue Hwang

February 2023

Thesis: A Study on Urban Disparities in Shared Bicycle Services Using Open Source Analysis Tools (In Korean)

- Built a Python package computing 2SFCA accessibility via Dijkstra routing with DEM-derived, slope-dependent cycling speeds. Validated travel-time estimates against GPS records and quantified building-level bike-share accessibility disparities across Seoul.

KYUNG HEE UNIVERSITY

Seoul, Korea

B.S. in Geography & B.A. in Tourism (Dual Degree, GPA 3.84/4.5)

February 2020

SELECTED PUBLICATIONS

Ji, S., Kim, S., Lee, J. & Seo, K. (2025). Quantitative Evaluation and Typology of Social Exposure Patterns to Urban Green Spaces: A Case Study of Seoul. *Forests*, 16(3), 510. <https://doi.org/10.3390/f16030510>

- Quantified demographic and temporal disparities in urban ecosystem-service accessibility via G2SFCA on de facto population and road-network data.

Choi, M., Jo, J., Kim, H. & Ji, S. (2025). Mapping the Unseen: Integrating Hiking Activities into Spatially Explicit Cultural Ecosystem Service Assessment Through Agent-Based Modeling. *Transactions in GIS*, 29(1), e13285. <https://doi.org/10.1111/tgis.13285>

- Contributed simulation-environment development and data visualization to an agent-based modeling (ABM) assessment of cultural ecosystem services from hiking trails.

Ji, S., Hwang, H., Hwang, C., Koo, C. & Lee, E. (2022). The Attributes of Virtual Reality (VR) Experiences and Spatial Cognition among Tourists: Focusing on a Behavioral Geographic Perspective. *Journal of Tourism Sciences*, 46(4), 153-176. <http://dx.doi.org/10.17086/JTS.2022.46.4.153.176> (In Korean)

- Reconstructed a campus-scale 3D environment from UAV imagery via SfM (Structure from Motion) photogrammetry and implemented it in Unity as an HMD-based VR stimulus for the experiment.

Ji, S., Seong, J., Stanescu, A., Hwang, C. & Lee, Y. (2021). Spatiotemporal Traffic Database Construction with Google Real-time Traffic Information and Congestion Pattern Analysis: A Case Study of Montgomery County, Maryland, USA. *Journal of the Korean Geographical Society*, 56(3), 265-276. <https://doi.org/10.22776/kgs.2021.56.3.265> (In Korean)

- Detected recurrent congestion patterns in a 3D spatiotemporal traffic database constructed from Google Maps real-time traffic layers.

RESEARCH EXPERIENCE

GEORGE MASON UNIVERSITY KOREA

Research Assistant (Remote) - Advisor: Dr. Sohyun Park

Incheon, Korea

2026 - Present

Explainable-AI-Based Analysis of Cropland Abandonment Trajectories

April 2026 - Present

- Typologizing post-abandonment cropland trajectories in Ohio from USDA Cropland Data Layer time series and interpreting their agro-ecological, demographic, and hazard-related drivers with explainable AI.

NATIONAL INSTITUTE OF FOREST SCIENCE

Research Assistant - Advisors: Dr. Sol-E Choi and Dr. Jeonghee Lee

Seoul, Korea

2023 - 2024, 2026

Development of Data Fusion Technologies for Terrestrial Resource Monitoring

April - August 2026

- Contributed to pre-launch preparation for Korea's first agriculture and forestry satellite (CAS500-4, launched July 2026), evaluating U-Net segmentation models and refining preprocessing for land-cover classification.
- Acquired UAV imagery of forest flux sites in field surveys for SfM 3D reconstruction and investigated UAV LiDAR point-cloud methods for forest resource assessment.

Assessment of Spatial Accessibility to Urban Ecosystems

December 2023 - September 2024

- Assessed urban green-space accessibility with G2SFCA on de facto population data and road networks.

KOREA RESEARCH INSTITUTE FOR LOCAL ADMINISTRATION

Research Assistant - Advisor: Dr. Wondo Lee

Wonju, Korea

2025 - 2026

Assessment of Energy Grid Strategies for Regional Equity

October 2025 - March 2026

- Built a nationwide power-grid network dataset from OpenStreetMap and satellite imagery for a national assessment of grid development and regional equity.

Analysis of the Daily Population Dynamics in Depopulating Areas

March 2025 - October 2025

- Identified shrinking regions nationwide using demographic and socioeconomic indicators and delineated neighborhood areas via community detection and BiFlowLISA on inter-regional flow networks.
- Processed de facto population and credit-card transaction data with SAS and SQL and examined visiting patterns in depopulating regions for a national policy report.

KYUNG HEE UNIVERSITY

Research Assistant - Advisor: Dr. Chulsue Hwang

Seoul, Korea

2020 - 2024

UAV/LiDAR-Based 3D Geospatial Modeling and Simulation

May 2022 - December 2024

Funded by the Korea Spatial Information Industry Promotion Agency (SpaceN)

- Processed UAV LiDAR point clouds and produced semantic 3D models through a Python reconstruction workflow and simulated seasonal energy demand and photovoltaic potential.

National Atlas of Korea Publication Project

July 2020 - December 2024

Funded by the Korean Geographical Society (KGS)

- Designed automated cartographic workflows for the National Atlas of Korea using ArcGIS ModelBuilder.

Satellite-Based Monitoring of Agriculture & Infrastructure in North Korea June 2020 - December 2022
Funded by the Korean Geographical Society (KGS)

- Quantified seasonal changes in crop-growing and reclaimed areas using satellite-derived NDVI and land-classification indices for food-security and disaster-risk assessment.
- Digitized roads, railways, and hubs from satellite imagery into a transportation-network geospatial database and assessed regional connectivity through GIS-based topological analysis.

Spatial Data Collection of Renewable Energy Facilities June 2020 - September 2020
Funded by the Korea Development Institute (KDI)

- Assembled a nationwide spatial panel dataset of renewable-energy facilities by integrating satellite-detected photovoltaic sites with public big-data sources.

UNIVERSITY OF WEST GEORGIA

Carrollton, GA, USA

Visiting Scholar - Advisor: Dr. Jeong Seong

2020 - 2022

Development of Analytical Methods for 3D Spatiotemporal Big Data Applying Machine Learning

June 2020 - May 2022

- Selected for a global training program and completed a nine-month research residency at UWG.
- Constructed 3D spatiotemporal traffic databases from Google Maps real-time traffic layers and classified recurrent and non-recurrent congestion with unsupervised clustering (DBSCAN, K-means).

TEACHING & MENTORING EXPERIENCE

KOREAN GEOGRAPHICAL SOCIETY

Seoul, Korea

Guest Lecturer

2024

Applications of Geographic Information Technologies

November - December 2024

- Lectured on 3D spatial-information construction and web-mapping services.

Geospatial Analysis and Applications Training Program

November 2024

- Lectured on vector, raster, and network data processing with QGIS.

KYUNG HEE UNIVERSITY

Seoul, Korea

Teaching Assistant

2020 - 2024

Applications of Geographic Information and Technology (GEOG4062)

Fall 2022 - 2024

- Developed and delivered the UAV practicum modules (SfM-based 3D reconstruction).

Geospatial Workshop, University of West Georgia

Summer 2023

- Led a hands-on geospatial analysis workshop for undergraduate students.

Introduction to Geographic Information Systems and Science (GEOG201401)

Spring 2020 - 2022

- Produced tutorial videos for GIS practical sessions and mentored undergraduate research projects.

Cartography (GEOG1007)

Fall 2020

- Monitored student progress and provided feedback on individual mapping projects.

HONORS & FUNDING

Kyung Hee University Academic Excellence Scholarship

July 2020 - February 2022

- Grant: \$3,500, Sponsored by Office of Educational Innovation & Planning, Kyung Hee University
- Awarded for research output (2 first-authored journal articles, 4 first-authored conference presentations)

Kyung Hee University Faculty-Nominated Research Scholarship

March 2021 - August 2021

- Grant: \$1,500, Nominated by Dr. Chulsue Hwang, Department of Geography, Kyung Hee University

High-Potential Individuals Global Training Program

October 2020 - June 2021

- Grant: \$15,900, Sponsored by Ministry of Science and ICT, Republic of Korea
- Visited University of West Georgia, GA as a visiting scholar (J-1 visa)

PROFESSIONAL DEVELOPMENT

Geospatial Professional Development

- NASA Applied Remote Sensing Training (ARSET) certificate of completion: Using Google Earth Engine for Land Monitoring Applications, June 2021
- Mentee, Korean Association of Geographic Information and Environmental Science (KAGES) Mentoring Program, Mentor: Dr. Taejin Park, NASA Ames Research Center, 2024
- Mentee, Korean Association of Geographic Information and Environmental Science (KAGES) Mentoring Program, Mentor: Dr. Junghwan Kim, Virginia Tech, 2022
- Research Assistant Intern, Origin-Destination (OD) Network Analysis Team, Department of National Transport Big Data, The Korea Transport Institute (KOTI), July - August 2018

Certification & Service

- Completed Curriculum for Teaching Profession in Geography and Tourism and Student Teaching Practicum (Seoul Tourism High School, May 2019)
- Military Training Instructor, Republic of Korea Air Force Basic Military Training Wing, 2014 - 2016

SKILLS

Programming & Scripting - Python, R, SQL, Bash (Ubuntu)

GIS & Remote Sensing Platforms - ArcGIS, QGIS, Google Earth Engine

3D Reconstruction & Visualization - Pix4D, Metashape, Trimble RealWorks, SketchUp, Unity, Meta Quest

UAV Platforms & Mission Planning - DJI Mavic, DJI Avata, DroneLink

LANGUAGE

Korean (native) • **English** (IELTS 7.5; academic writing, conference presentations, international collaboration)

SELECTED CONFERENCE PRESENTATIONS

International Presentations

Kim, J., Nam, H., **Ji, S.** & Hwang, C. (2024). Construction of CityGML-Based 3D Digital Twin using UAV: A Case Study of Kyung Hee University, Seoul. In *Proceedings of the 2024 American Association of Geographers Annual Meeting*, April 2024, Honolulu, USA. ([Abstract](#))

Ji, S., Kim, J., Nam, H. & Lee, J. (2024). Assessing Spatial Accessibility in Forest Welfare for Enhanced Quality of Life. In *Proceedings of the 2024 American Association of Geographers Annual Meeting*, April 2024, Honolulu, USA. ([Abstract](#), [Presentation](#), [Related Article](#))

Choi, M., Jo, J., Kim, H. & **Ji, S.** (2023). Assessing Cultural Ecosystem Services Supply and Demand using Agent-Based Modeling. *The International Symposium on Sustainable Forest Ecosystem Management*, August 2023, Jeju, Korea. ([Presentation](#), [Related Article](#))

Kim, H., Cho, S., Hong, H., Seong, J., Stanescu, A., Hwang, C. & **Ji, S.** (2022). Traffic Big-data Analysis with Google Maps Spatio-temporal Database: Atlanta, Georgia, 2021. In *Proceedings of the 2022 American Association of Geographers Annual Meeting*, February 2022, Virtual meeting. ([Abstract](#))

Ji, S., Seong, J., Lee, Y., Stanescu, A. & Hwang, C. (2021). An Exploratory Spatial Pattern Analysis Using a Traffic Congestion Data-Mining Approach with an Online Map Service. In *Proceedings of the 2021 American Association of Geographers Annual Meeting*, April 2021, Virtual meeting. ([Abstract](#), [Presentation](#), [Related Article](#))

Domestic Presentations

Kang, J., Park, J., **Ji, S.**, Lee, J. & Seo, K. (2024). Modifiable Areal Unit Problem in Measuring Spatial Accessibility to Urban Green Space: The Case Study of Seoul. In *Proceedings of the 2024 Summer Conference of the Korean Society of Forest Science*, August 2024, Jeju, Korea. ([Abstract](#))

Ji, S., Lee, S., Hwang, T. & Lee, W. (2022). Open Source Package for Measuring Geographic Accessibility using the National Traffic Information Network. In *Proceedings of the International Conference of the Korean Society of Transportation*, September 2022, Seoul, Korea. ([Abstract](#), [Presentation](#), [Related Article](#))

Posters & Others

Shin, E., **Ji, S.**, Kim, S., Kim, E., Park, C. & Choi, S. (2026). Nationwide Land Cover Classification Using Multi-Temporal Sentinel-2 Imagery and Deep Learning. In *Proceedings of the 2026 Annual Conference of the Korean Society of Remote Sensing*, October 2026, Busan, Korea. (Accepted)

Ji, S., Lee, J. & Seo, K. (2024). Exploring Population Exposure Patterns to Urban Green Spaces Using the 2SFCA Methodology. In *Proceedings of the World Forum on Forest Therapy 2024*, October 2024, Daejeon, Korea. ([Abstract](#), [Poster](#))

Ji, S., Lee, J. & Seo, K. (2024). Quantifying Urban Green Space Accessibility in Seoul: Applying the Gaussian 2-Step Floating Catchment Area (G2SFCA). In *Proceedings of the 2024 Summer Conference of the Korean Society of Forest Science*, August 2024, Jeju, Korea. ([Abstract](#), [Poster](#))

Ji, S., Lee, E. & Hwang, H. (2022). Development of Immersive Virtual Reality from a Pedestrian Perspective Using a UAV-SfM 3D Model. In *Proceedings of the 1st Kyung Hee University Graduate School Joint Academic Seminar*, January 2022, Seoul, Korea. ([Abstract](#), [Poster](#), [Related Article](#)). ISBN: 979-11-977927-0-0.

Ji, S., Seong, J., Stanescu, A., Hwang, C. & Lee, Y. (2021). Exploratory Spatiotemporal Congestion Pattern Analysis Using Real-time Traffic Congestion Data Collection Method within Online Map Service: A Case Study of Seoul. In *Proceedings of the 2021 Annual Conference of the Korean Geographical Society*, June 2021, Virtual meeting. ([Abstract](#), [Poster](#), [Related Article](#))

Seong, J., **Ji, S.**, Stanescu, A., Lee, Y. & Hwang, C. (2021). What will be Presented at the 2021 AAG Meeting? *AAG Newsletter*, 55, April 2021. ([Article](#), [Related Article](#))